Enstrophy and Weighted-Enstrophy Balance Identities on $\mathbb{T}^3$: a Lean 4 Formalization

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Abstract

We formalize in Lean 4 with Mathlib two exact balance identities from the a-posteriori Lyapunov functional approach to the 3-D incompressible Navier–Stokes vorticity equation on the torus: the enstrophy identity $\frac{dE}{dt} = P - \nu \int |\nabla\omega|^2$ and the weighted identity for $Q = \int |\omega|^4\phi$. Both are proved for every periodic differential algebra whose integral annihilates spatial derivatives and commutes with the time derivative, for trigonometric polynomials on $\mathbb{T}^3$, and for smooth (C^∞) fields on $\mathbb{R}\times\mathbb{T}^3$. We also formalize the L^2 strain identity, the exact evolution of $S = E - \frac{\kappa}{2}Q$, a weight-along-paths bound, and boundedness of the comparison ODE. The development has 0 `sorry` and uses only the axioms `propext`, `Classical.choice`, `Quot.sound`.

1 Introduction

Commentary (adapted from the author's paper). Classical energy methods for the Navier–Stokes equations control the kinetic energy but not the enstrophy in three dimensions; the vortex-stretching term can produce growth that is difficult to absorb. The approach studies the exact classical equations together with an auxiliary scalar field ϕ transported by $|\omega|^2$, and the two integrals $E = \frac{1}{2}\int |\omega|^2$ and $Q = \int |\omega|^4\phi$. The first step is to compute dE/dt and dQ/dt exactly. Those two computations are what this paper formalizes. They are identities: no inequality and no bound on the solution is involved.

2 Setting

Let $u, \omega : \mathbb{T}^3 \to \mathbb{R}^3$ and $\phi : \mathbb{T}^3 \to \mathbb{R}$ depend on t, with

$$\partial_k u_k = 0, \tag{1}$$
$$\partial_t \omega_i = -u_k \partial_k \omega_i + \omega_k \partial_k u_i + \nu\, \partial_k \partial_k \omega_i, \tag{2}$$
$$\partial_t \phi = -u_k \partial_k \phi + |\omega|^2, \tag{3}$$

summing over repeated indices. The relation $\omega = \nabla \times u$ is not needed for either identity.

Definition 1 (Periodic setting, `PeriodicSetting`)**.** A commutative K-algebra A with K-derivations $\partial_1, \partial_2, \partial_3, \partial_t$, a K-module B, a K-linear map $I : A \to B$ and a map $D_t : B \to B$ such that $I(\partial_k f) = 0$ for all k, f and $I(\partial_t f) = D_t(If)$.

Section 6 proves both axioms for smooth periodic fields.

Definition 2 (Trigonometric model, `Trig.setting`)**.** $A = R[\mathbb{Z}^3]$ (finite sums $\sum_n f_n e^{2\pi i n\cdot x}$, $f_n \in R$), where R is a commutative $\mathbb{C}$-algebra with a derivation D. Set $\partial_k(\sum f_n e_n) = \sum 2\pi i n_k f_n e_n$, $\partial_t(\sum f_n e_n) = \sum D(f_n)e_n$, $I(f) = f_0$, $D_t = D$.

Lemma 3 (`dx`, `dt`)**.** *∂_k and ∂_t are derivations of $R[\mathbb{Z}^3]$.*

Proof. Both are linear, so the Leibniz rule reduces to monomials. For ∂_k: $\partial_k(ae_n \cdot be_m) = 2\pi i(n_k + m_k)ab\, e_{n+m} = ae_n \cdot 2\pi i m_k b e_m + be_m \cdot 2\pi i n_k a e_n$. For ∂_t: $D(ab)e_{n+m} = (aDb + bDa)e_{n+m}$. □

Lemma 4 (Torus divergence lemma, `integral_dx`, `integral_dt`)**.** *$I(\partial_k f) = 2\pi i \cdot 0 \cdot f_0 = 0$ and $I(\partial_t f) = D(f_0) = D(If)$.*

Hence the trigonometric model is a periodic setting, and every theorem below holds for it.

3 Pointwise lemmas

Lemma 5 (`adv_eq`)**.** *$u_k\partial_k g = \partial_k(u_k g) - (\partial_k u_k)g$. Hence (`int_adv_eq_zero`) if (1) holds, $I(u_k\partial_k g) = 0$.*

Proof. Leibniz: $\partial_k(u_k g) = u_k\partial_k g + g\partial_k u_k$. Apply I and use $I \circ \partial_k = 0$ and (1). □

Lemma 6 (`wLapW_eq`)**.** *$\omega_i\partial_k\partial_k\omega_i = \partial_k(\omega_i\partial_k\omega_i) - \partial_k\omega_i\,\partial_k\omega_i$.*

Proof. Leibniz on $\partial_k(\omega_i \cdot \partial_k\omega_i)$. □

Lemma 7 (`two_w_adv`)**.** *$2\,\omega_i\,u_k\partial_k\omega_i = u_k\partial_k|\omega|^2$.*

Proof. $\partial_k(\omega_i\omega_i) = 2\omega_i\partial_k\omega_i$. □

Lemma 8 (`stretch_eq`)**.** *$\omega_i\omega_j\partial_j u_i = \omega \cdot (\omega \cdot \nabla)u$.*

4 Identity (1)

Lemma 9 (`dt_half_sq`, `dt_vsq_div`)**.** *Under (1), (2):*

$$\partial_t|\omega|^2 = 2\,\omega_i\omega_j\partial_j u_i - 2\nu\,|\nabla\omega|^2 + \partial_k\big(2\nu\,\omega_i\partial_k\omega_i - u_k|\omega|^2\big).$$

Proof. $\partial_t|\omega|^2 = 2\omega_i\partial_t\omega_i$. Substitute (2): $2\omega_i\partial_t\omega_i = -2\omega_i u_k\partial_k\omega_i + 2\omega_i\omega_k\partial_k u_i + 2\nu\,\omega_i\partial_k\partial_k\omega_i$. By `two_w_adv` and `adv_eq` with (1), the first term is $-\partial_k(u_k|\omega|^2)$. The second is $2\omega_i\omega_j\partial_j u_i$. By `wLapW_eq`, the third is $2\nu\,\partial_k(\omega_i\partial_k\omega_i) - 2\nu|\nabla\omega|^2$. □

Commentary. The transport term vanishes by periodicity and $\nabla \cdot u = 0$; the viscous term integrates by parts to $-\nu\int|\nabla\omega|^2$; the stretching term remains.

Theorem 10 (Identity (1), `enstrophy_identity`, `enstrophy_identity_T3`)**.** *With $E = \frac{1}{2}\int|\omega|^2$ and $P = \int\omega_i\omega_j\partial_j u_i$,*

$$\frac{dE}{dt} = P - \nu\int_{\mathbb{T}^3}|\nabla\omega|^2.$$

(Formalized as $D_t\,I(|\omega|^2) = 2\,I(\omega_i\omega_j\partial_j u_i) - 2\nu\,I(|\nabla\omega|^2)$.)

Proof. $D_t I(|\omega|^2) = I(\partial_t|\omega|^2)$; apply I to the lemma; the divergence term has $I = 0$. □

5 Identity (2)

Write $s = |\omega|^2$, so $|\omega|^4\phi = s^2\phi$.

Lemma 11 (`dt_weighted`)**.** *Under (2), (3):*

$$\partial_t(s^2\phi) = 4s\phi\,\omega_i\omega_j\partial_j u_i + 4\nu\, s\phi\,\omega\cdot\Delta\omega + s^3 - u_k\partial_k(s^2\phi).$$

Proof. $\partial_t(s^2\phi) = 2s\phi\,\partial_t s + s^2\partial_t\phi$. From the proof of (1), $\partial_t s = 2\omega_i\omega_j\partial_j u_i + 2\nu\,\omega\cdot\Delta\omega - u_k\partial_k s$, and by (3) $\partial_t\phi = -u_k\partial_k\phi + s$. So

$$\partial_t(s^2\phi) = 4s\phi\,\omega_i\omega_j\partial_j u_i + 4\nu s\phi\,\omega\cdot\Delta\omega - \big(2s\phi\,u_k\partial_k s + s^2 u_k\partial_k\phi\big) + s^3,$$

and $2s\phi\,u_k\partial_k s + s^2 u_k\partial_k\phi = u_k\partial_k(s^2\phi)$ by Leibniz. □

Commentary (adapted from the author's paper). The transport contributions cancel exactly. Using $\omega\cdot(u\cdot\nabla)\omega = \frac{1}{2}(u\cdot\nabla)|\omega|^2$ they combine into the pure divergence $-\int u\cdot\nabla(\phi|\omega|^4)$, which vanishes on $\mathbb{T}^3$.

Lemma 12 (`dt_weighted_div`)**.** *Under (1) as well,* $\partial_t(s^2\phi) = 4s\phi\,\omega_i\omega_j\partial_j u_i + 4\nu s\phi\,\omega\cdot\Delta\omega + s^3 - \partial_k(u_k s^2\phi)$.

Proof. `adv_eq` with $g = s^2\phi$ and (1). □

Theorem 13 (Identity (2), `weighted_identity`, `weighted_identity_T3`)**.** *With* $Q = \int|\omega|^4\phi$,

$$\frac{dQ}{dt} = 4\int|\omega|^2\phi\,\omega\cdot(\omega\cdot\nabla)u + 4\nu\int|\omega|^2\phi\,\omega\cdot\Delta\omega + \int|\omega|^6.$$

Proof. $D_t I(s^2\phi) = I(\partial_t(s^2\phi))$; apply I, the divergence has $I = 0$, and use `stretch_eq`. □

6 Smooth periodic fields

Definition 14 (`Smooth.A`, `Smooth.setting`)**.** A is the algebra of $f \in C^\infty(\mathbb{R}\times\mathbb{R}^3,\mathbb{R})$, $f = f(t,a,b,c)$, with $f(p+e_k) = f(p)$ for the unit spatial vectors e_1, e_2, e_3. $\partial_k f(p) = Df(p)e_k$, $\partial_t f(p) = Df(p)e_t$, $I(f)(t) = \int_0^1\int_0^1\int_0^1 f(t,a,b,c)\,dc\,db\,da$, $D_t = \frac{d}{dt}$.

Lemma 15 (`Torus.hasDerivAt_integral01`)**.** *If $F, \partial_s F$ are jointly continuous on $\mathbb{R}\times\mathbb{R}$, then* $\frac{d}{ds}\int_0^1 F(s,x)\,dx = \int_0^1 \partial_s F(s,x)\,dx$.

Proof. Dominated differentiation: $\partial_s F$ is bounded on the compact set $\bar{B}(s_0,1)\times[0,1]$. □

Lemma 16 (`Torus.hasDerivAt_torusIntegral`; (b))**.** *If $f, \partial_t f$ are jointly continuous, then* $\frac{d}{dt}\int_{\mathbb{T}^3} f = \int_{\mathbb{T}^3}\partial_t f$.

Proof. Apply the previous lemma three times; each inner integral is jointly continuous. □

Lemma 17 (`Torus.torusIntegral_deriv_a/b/c`; (a))**.** *If f is C^1 and 1-periodic in the differentiated variable, then* $\int_{\mathbb{T}^3}\partial_k f = 0$.

Proof. Differentiate the inner integrals in that variable using the first lemma, then apply $\int_0^1 g' = g(1) - g(0) = 0$. □

Theorem 18 (`Smooth.int_d`, `Smooth.int_dt`, `Smooth.enstrophy_identity_smooth`, `Smooth.weighted_identi`
The smooth model is a periodic setting. Hence identities (1) and (2) hold for all $u, \omega, \phi \in A$ satisfying (1)–(3).

7 Strain, the functional S, the weight, and the comparison ODE

Theorem 19 (`Fourier.L2_strain_eq`, `Fourier.L2_strain_le`)**.** *Take $u, \omega, S \in L^2(\mathbb{T}^3)$ whose Fourier coefficients are those of a divergence-free velocity, its curl, and $S = \frac{1}{2}(\nabla u + \nabla u^T)$. Then $\sum_{ij} \|S_{ij}\|_2^2 = \frac{1}{2}\sum_i \|\omega_i\|_2^2$, so $\|S\|_2 \le \|\omega\|_2$.*

Proof. Mode by mode, $\sum |\hat{S}_{ij}|^2 = \frac{1}{2}|\hat{\omega}|^2$ when $k \cdot \hat{u} = 0$. Then apply Parseval. □

Theorem 20 (`Lyapunov.dS_identity_smooth`)**.** *Let $S = \frac{1}{2}\int |\omega|^2 - \frac{\kappa}{2}\int |\omega|^4\phi$. For smooth periodic solutions of (1)–(3),*

$$S' = \int \omega_i\omega_j\partial_j u_i - \nu\int |\nabla\omega|^2 - \tfrac{\kappa}{2}\Big(4\int |\omega|^2\phi\,\omega_i\omega_j\partial_j u_i + 4\nu\int |\omega|^2\phi\,\omega\cdot\Delta\omega + \int |\omega|^6\Big).$$

Theorem 21 (`Lyapunov.le_integral_sq_of_deriv`)**.** *Let h, M be continuous, $g(0) = 0$, and $g' = h \le M^2$ on $[0,t]$. Then $g(t) \le \int_0^t M^2$.*

Take $g(s) = \phi(s, X(s))$ along a particle path. Then $g' = |\omega|^2(s, X(s))$, so $\phi \le \int_0^t M^2$ along paths. The chain rule step is not formalized.

Theorem 22 (`Lyapunov.ode_bounded`)**.** *Let $\kappa > 0$, and let $y' = y^2(C - \kappa y)$ on $[0,T]$. Then $\min(y_0, C/\kappa, 0) \le y(t) \le \max(y_0, C/\kappa, 0)$.*

Proof. Comparison with the barrier $B + \varepsilon$, where $B = \max(y_0, C/\kappa, 0)$: at $y = B + \varepsilon$ we have $y' < 0$. Then let $\varepsilon \to 0$. The lower bound follows in the same way. □

Stated, not proved. `CalderonZygmundLp` p is the bound $\int |\partial_j u_i|^p \le C\int |\omega|^p$; it is classical for $1 < p < \infty$ and is not formalized here. `WeightedToSupReduction` says a bound on S implies a bound on $\sup|\omega|^2$. `PhiSupControl` is a uniform bound on $|\phi|$. These are `Prop` definitions only, and no theorem uses them. The L^∞ Calderón–Zygmund bound is false and is not stated.

8 Scope

Identities (1) and (2) and the results of the last section are proved as stated. The absorption step that uses an L^∞ Calderón–Zygmund bound, the reduction from S to $\sup|\omega|$, and the bootstrap are not proved. The paper makes no regularity claim.

Verification

`lake build`: success, 0 `sorry`. `lake env lean AxiomAudit.lean`: only `propext`, `Classical.choice`, `Quot.sound`. Toolchain `leanprover/lean4:v4.30.0-rc1`, Mathlib `v4.30.0-rc1`.